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KELAS : TI-25-KA
MATKUL : LAB_ALGORITMA

1. Buatlah program perhitungan matriks (penjumlahan, pengurangan, dan perkalian) yang dapat diinput manual elemennya!

   Ini programnya:

```cpp
#include <iostream>
using namespace std;

int main() {
    int baris, kolom;

    cout << "PROGRAM OPERASI MATRIKS\n";
    cout << "=========================\n\n";

    cout << "Masukkan jumlah baris matriks : ";
    cin >> baris;
    cout << "Masukkan jumlah kolom matriks : ";
    cin >> kolom;

    int A[10][10], B[10][10], C[10][10];

    cout << "\nInput Matriks A:\n";
    for (int i = 0; i < baris; i++) {
        for (int j = 0; j < kolom; j++) {
            cout << "A[" << i << "][" << j << "] = ";
            cin >> A[i][j];
        }
    }

    cout << "\nInput Matriks B:\n";
    for (int i = 0; i < baris; i++) {
        for (int j = 0; j < kolom; j++) {
            cout << "B[" << i << "][" << j << "] = ";
            cin >> B[i][j];
        }
    }

    cout << "\nHASIL PENJUMLAHAN (A + B):\n";
    for (int i = 0; i < baris; i++) {
        for (int j = 0; j < kolom; j++) {
            C[i][j] = A[i][j] + B[i][j];
            cout << C[i][j] << "\t";
        }
        cout << endl;
    }

    cout << "\nHASIL PENGURANGAN (A - B):\n";
    for (int i = 0; i < baris; i++) {
        for (int j = 0; j < kolom; j++) {
            C[i][j] = A[i][j] - B[i][j];
            cout << C[i][j] << "\t";
        }
        cout << endl;
    }

    cout << "\nHASIL PERKALIAN MATRIKS (A x B):\n";
    for (int i = 0; i < baris; i++) {
        for (int j = 0; j < kolom; j++) {
            C[i][j] = 0;
            for (int k = 0; k < kolom; k++) {
                C[i][j] += A[i][k] * B[k][j];
            }
            cout << C[i][j] << "\t";
        }
        cout << endl;
    }

    return 0;
}
```

Ini programnya ketika dijalankan:

```
PROGRAM OPERASI MATRIKS
==========================

Masukkan jumlah baris matriks : 3
Masukkan jumlah kolom matriks : 3

Input Matriks A:
A[0][0] = 10
A[0][1] = 5
A[0][2] = 9
A[1][0] = 12
A[1][1] = 13
A[1][2] = 5
A[2][0] = 14
A[2][1] = 4
A[2][2] = 12

Input Matriks B:
B[0][0] = 6
B[0][1] = 8
B[0][2] = 10
B[1][0] = 23
B[1][1] = 11
B[1][2] = 7
B[2][0] = 9
B[2][1] = 2
B[2][2] = 18

HASIL PENJUMLAHAN (A + B):
16      13      19
35      24      12
23      6       30

HASIL PENGURANGAN (A - B):
4       -3      -1
-11     2       -2
5       2       -6

HASIL PERKALIAN MATRIKS (A x B):
256     153     297
416     249     301
284     180     384

--------------------------------
Process exited after 210.7 seconds with return value 0
Press any key to continue . . .
```

2. Sebuah perusahaan ayam goreng dengan nama “GEROBAK FRIED CHICKEN” yang telah lumayan banyak pelanggannya, ingin dibantu dibuatkan program untuk membantu kelancaran usahanya. “GEROBAK FRIED CHICKEN” mempunyai daftar harga ayam sebagai berikut:

   Kode — Jenis — Harga
   D — Dada — Rp. 2500
   P — Paha — Rp. 2000
   S — Sayap — Rp. 1500

   Ini programnya:

```cpp
#include <iostream>
#include <iomanip>
using namespace std;

int main() {
    int banyakJenis;
    cout << "GEROBAK FRIED CHICKEN\n";
    cout << "==============================\n";
    cout << "Kode   Jenis    Harga\n";
    cout << "==============================\n";
    cout << "D      Dada     Rp.2500\n";
    cout << "P      Paha     Rp.2000\n";
    cout << "S      Sayap    Rp.1500\n";
    cout << "==============================\n\n";

    cout << "Banyak Jenis : ";
    cin >> banyakJenis;

    char kode[20];
    int banyakPotong[20];
    int hargaSatuan[20];
    string namaJenis[20];
    int jumlahHarga[20];

    for (int i = 0; i < banyakJenis; i++) {
        cout << "\nJenis ke-" << i + 1 << endl;
        cout << "Jenis Potong [D/P/S] : ";
        cin >> kode[i];

        if (kode[i] == 'D' || kode[i] == 'd') {
            hargaSatuan[i] = 2500;
            namaJenis[i] = "Dada";
        }
        else if (kode[i] == 'P' || kode[i] == 'p') {
            hargaSatuan[i] = 2000;
            namaJenis[i] = "Paha";
        }
        else if (kode[i] == 'S' || kode[i] == 's') {
            hargaSatuan[i] = 1500;
            namaJenis[i] = "Sayap";
        }
        else {
            cout << "Kode tidak valid! otomatis dianggap Sayap.\n";
            hargaSatuan[i] = 1500;
            namaJenis[i] = "Sayap";
        }

        cout << "Banyak Potong : ";
        cin >> banyakPotong[i];

        jumlahHarga[i] = hargaSatuan[i] * banyakPotong[i];
    }

    cout << "\n============================================================\n";
    cout << "                    GEROBAK FRIED CHICKEN\n";
    cout << "============================================================\n";
    cout << left << setw(5) << "No"
         << setw(12) << "Jenis"
         << setw(12) << "Harga"
         << setw(10) << "Banyak"
         << "Jumlah Harga\n";
    cout << "------------------------------------------------------------\n";

    int totalBayar = 0;
    for (int i = 0; i < banyakJenis; i++) {
        cout << left << setw(5) << i + 1;
        cout << setw(12) << namaJenis[i];
        cout << "Rp." << setw(8) << hargaSatuan[i];
        cout << setw(10) << banyakPotong[i];
        cout << "Rp." << jumlahHarga[i] << endl;

        totalBayar += jumlahHarga[i];
    }

    float pajak = totalBayar * 0.10;
    float totalAkhir = totalBayar + pajak;

    cout << "--------------------------------------------\n";
    cout << "Jumlah Bayar\t: Rp. " << totalBayar << endl;
    cout << "Pajak 10%\t: Rp. " << pajak << endl;
    cout << "Total Bayar\t: Rp. " << totalAkhir << endl;
    cout << "--------------------------------------------\n";

    return 0;
}
```

Ini programnya ketika dijalankan:

```
GEROBAK FRIED CHICKEN
=============================
Kode   Jenis    Harga
=============================
D      Dada     Rp.2500
P      Paha     Rp.2000
S      Sayap    Rp.1500
=============================

Banyak Jenis : 3

Jenis ke-1
Jenis Potong [D/P/S] : p
Banyak Potong : 10

Jenis ke-2
Jenis Potong [D/P/S] : d
Banyak Potong : 10

Jenis ke-3
Jenis Potong [D/P/S] : s
Banyak Potong : 10

==============================================================
                    GEROBAK FRIED CHICKEN
==============================================================
No   Jenis       Harga       Banyak    Jumlah Harga
--------------------------------------------------------------
1    Paha        Rp.2000    10        Rp.20000
2    Dada        Rp.2500    10        Rp.25000
3    Sayap       Rp.1500    10        Rp.15000
-----------------------------------------------
Jumlah Bayar    : Rp. 60000
Pajak 10%       : Rp. 6000
Total Bayar     : Rp. 66000
-----------------------------------------------


--------------------------------
Process exited after 18.67 seconds with return value 0
Press any key to continue . . .
```